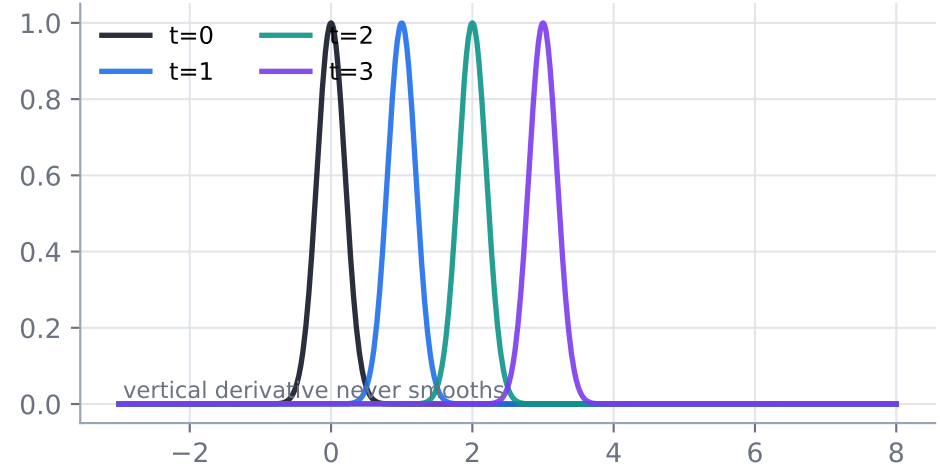
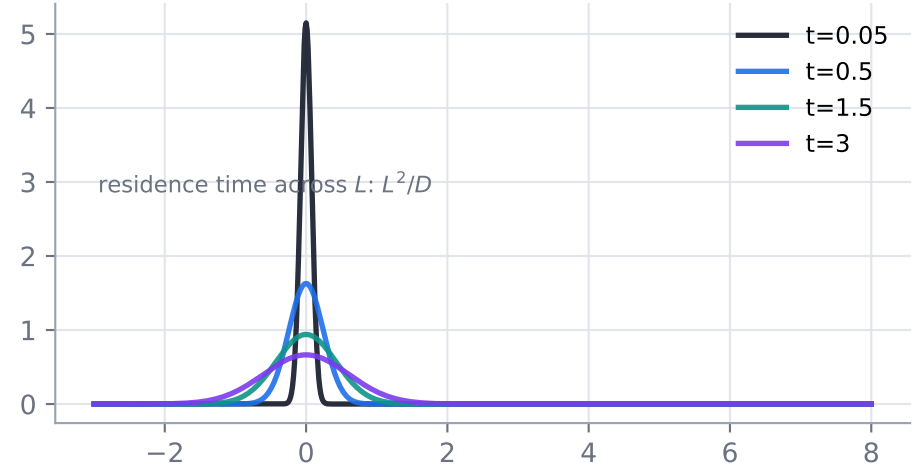
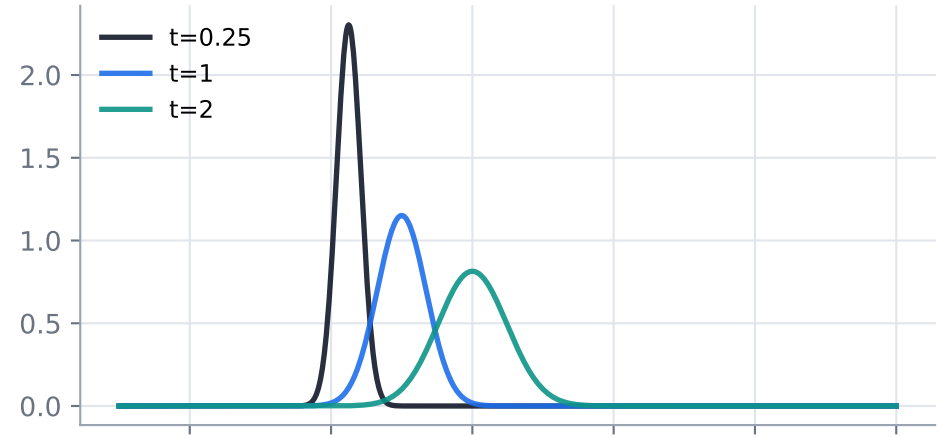
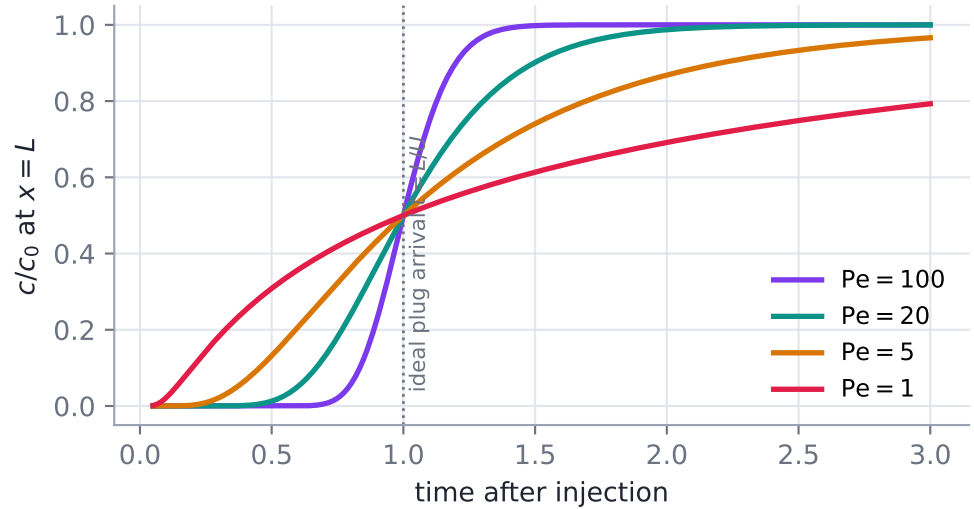
pure advection: $\partial_t c + U\partial_x c = 0$
the shape is frozen; only the axis moves



pure diffusion: $\partial_t c = D\partial_x^2 c$
area is conserved, peak decays as $t^{-1/2}$, width grows as $t^{1/2}$



advection + diffusion (Pe $= UL/D \approx 17$)
the two terms commute here — but not once D or U vary low Pe: smeared and early; high Pe: sharp and plug-like



breakthrough curve at the outlet